New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers. New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.
As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Data Description

The data provided has

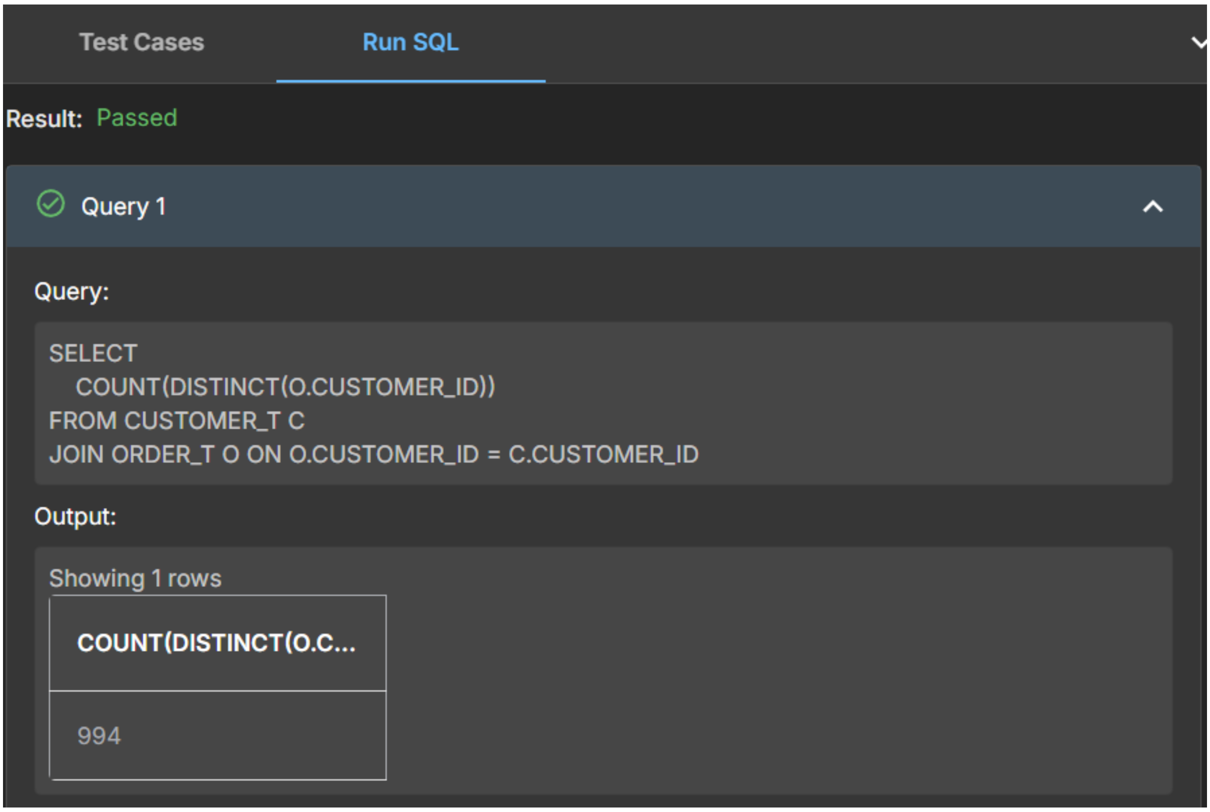
- Attributes on the vehicles New-Wheels sells - What are the make, model, and year?
- What is the price point?
- Attributes on the customers, such as where they live and payment methods
- Attributes on orders and shipments, such as when the order was shipped and received, what the after-sales feedback was, and so on.

BUSINESS QUESTION

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT
   COUNT (DISTINCT (O. CUSTOMER_ID))
FROM CUSTOMER_T C
JOIN ORDER_T O ON O. CUSTOMER_ID = C. CUSTOMER_ID;
```



OBSERVATION:

- Total customer who placed orders are 994.

```
SELECT
    STATE,
    COUNT(CUSTOMER_ID) AS TOTAL_CUSTOMER
FROM CUSTOMER_T
GROUP BY STATE;
```

Query 2

Query:

```
SELECT
   STATE,
   COUNT(CUSTOMER_ID) AS TOTAL_CUSTOMER
FROM CUSTOMER_T
GROUP BY STATE
```

Output:

Showing first 10 rows out of 49 rows

state	TOTAL_CUSTOMER
Alabama	29
Alaska	10
Arizona	26
Arkansas	6

OBSERVATION:

- There is uneven distribution.
- Customer count varies significantly across states.
- Some states have high concentration (California- 97 customers, Florida – 86 customers).
- Some state shows moderate customer presence (District of Columbia, Colorado-33, Alabama -29, Arizona -26, Connecticut -22)
- Some states have low customer (Arkansas-6, Delaware-6, Alaska-10).

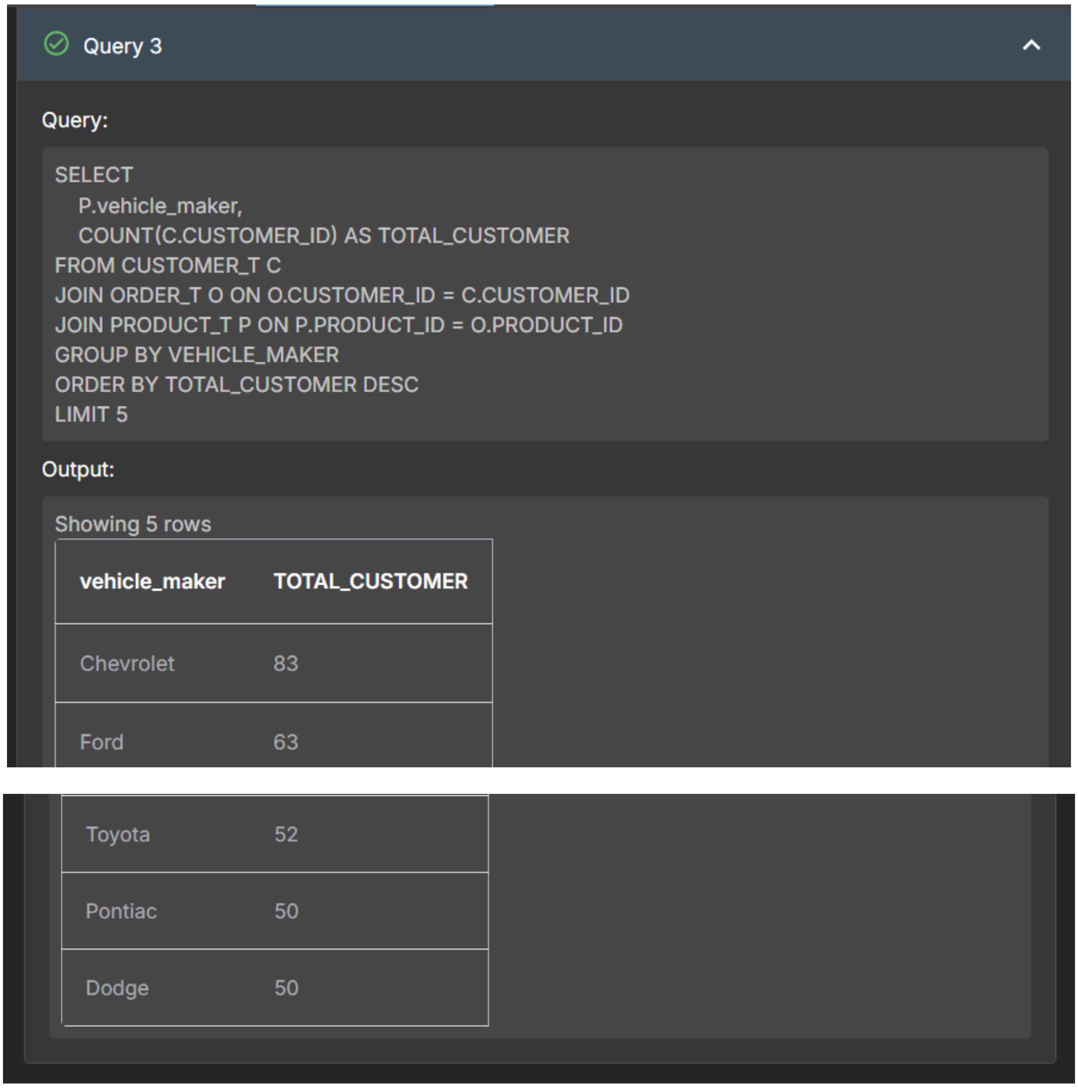
INSIGHT:

- 994 unique customers have made at least one purchase.
- Indicates a large active customer base.
- This Suggests good customer engagement and conversion.
- The distribution across the state is uneven, states like California Florida has highest number of customers indicating strong market presence. Moderate contribution is seen from state like Columbia Colorado and Alabama.
- However, states such as Arkansas, Delaware, and Alaska have very low customer counts, highlighting potential areas for business expansion and targeted marketing strategies.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

SOLUTION QUERY:

```
SELECT
    P.vehicle_maker,
    COUNT (C. CUSTOMER_ID) AS TOTAL_CUSTOMER
FROM CUSTOMER_T C
JOIN ORDER_T O ON O.CUSTOMER_ID = C.CUSTOMER_ID
JOIN PRODUCT_T P ON P.PRODUCT_ID = O.PRODUCT_ID
GROUP BY VEHICLE_MAKER
ORDER BY TOTAL_CUSTOMER DESC
LIMIT 5;
```



Query 3

Query:

```
SELECT
    P.vehicle_maker,
    COUNT(C.CUSTOMER_ID) AS TOTAL_CUSTOMER
FROM CUSTOMER_T C
JOIN ORDER_T O ON O.CUSTOMER_ID = C.CUSTOMER_ID
JOIN PRODUCT_T P ON P.PRODUCT_ID = O.PRODUCT_ID
GROUP BY VEHICLE_MAKER
ORDER BY TOTAL_CUSTOMER DESC
LIMIT 5
```

Output:

Showing 5 rows

vehicle_maker	TOTAL_CUSTOMER
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

OBSERVATION:

- Most popular brand preferred by customer Chevrolet.
- Ford and Toyota this brand also have strong customer demands.
- Pontiac and Dodge have equal preference.

INSIGHT:

- Chevrolet dominates customer preference, indicating strong brand loyalty and higher market demand.
- Ford and Toyota are strong competitors, showing consistent customer interest and competitive positioning.
- The difference between Toyota (52) and Pontiac/Dodge (50) is small, suggesting tight competition among mid-tier brands.
- Overall, the market shows strong competition with a few dominant players and minimal differences among the top five vehicle makers.

Question 3: Which is the most preferred vehicle maker in each state?

SOLUTION QUERY:

```
SELECT
 STATE, VEHICLE_MAKER, TOTAL_CUSTOMER
FROM
 (
  SELECT
        C.STATE, P. VEHICLE_MAKER,
        COUNT(C.CUSTOMER_ID) AS TOTAL_CUSTOMER,
 RANK () OVER (PARTITION BY C. STATE ORDER BY
 COUNT (C. CUSTOMER_ID) DESC) AS RNK
FROM CUSTOMER_T C
JOIN ORDER_T O ON O. CUSTOMER_ID = C. CUSTOMER_ID
JOIN PRODUCT_T P ON P.PRODUCT_ID = O.PRODUCT_ID
GROUP BY C. STATE, P. VEHICLE_MAKER
) AS TEMP
WHERE RNK=1;
```

Query 4

Query:

```
SELECT
   STATE,VEHICLE_MAKER,TOTAL_CUSTOMER
FROM
   (
      SELECT
   C.STATE,P.VEHICLE_MAKER,
   COUNT(C.CUSTOMER_ID) AS TOTAL_CUSTOMER,
   RANK() OVER(PARTITION BY C.STATE ORDER BY COUNT(C.CUSTOMER_ID)DESC) AS RNK

FROM CUSTOMER_T C
JOIN ORDER_T O ON O.CUSTOMER_ID = C.CUSTOMER_ID
JOIN PRODUCT_T P ON P.PRODUCT_ID = O.PRODUCT_ID
GROUP BY C.STATE,P.VEHICLE_MAKER

) AS TEMP
WHERE RNK=1
```

Output:

Output:

Showing first 10 rows out of 143 rows

STATE	VEHICLE_MAKER	TOTAL_CUSTOMER
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Volkswagen	1
Arkansas	Suzuki	1
Arkansas	Pontiac	1

OBSERVATION:

- Each state has its own preferred vehicle maker, no single brand dominates all state.
- In Some states, multiple brands have the same highest count. This is because we have used rank function which allow ties.
- Some states have very small demands which indicates weak demands.

INSIGHT:

- Customer preferences vary significantly across states, indicating that vehicle demand is influenced by regional factors such as availability, pricing, and customer behaviour

- There is no single vehicle maker dominating all states, suggesting a highly competitive and fragmented market.
- In states like Arizona and Arkansas there is no particular preference multiple brands have equal rank, indicating no clear customer preference and equal competitions.
- low customer counts in certain states suggest potential opportunities for business expansion.

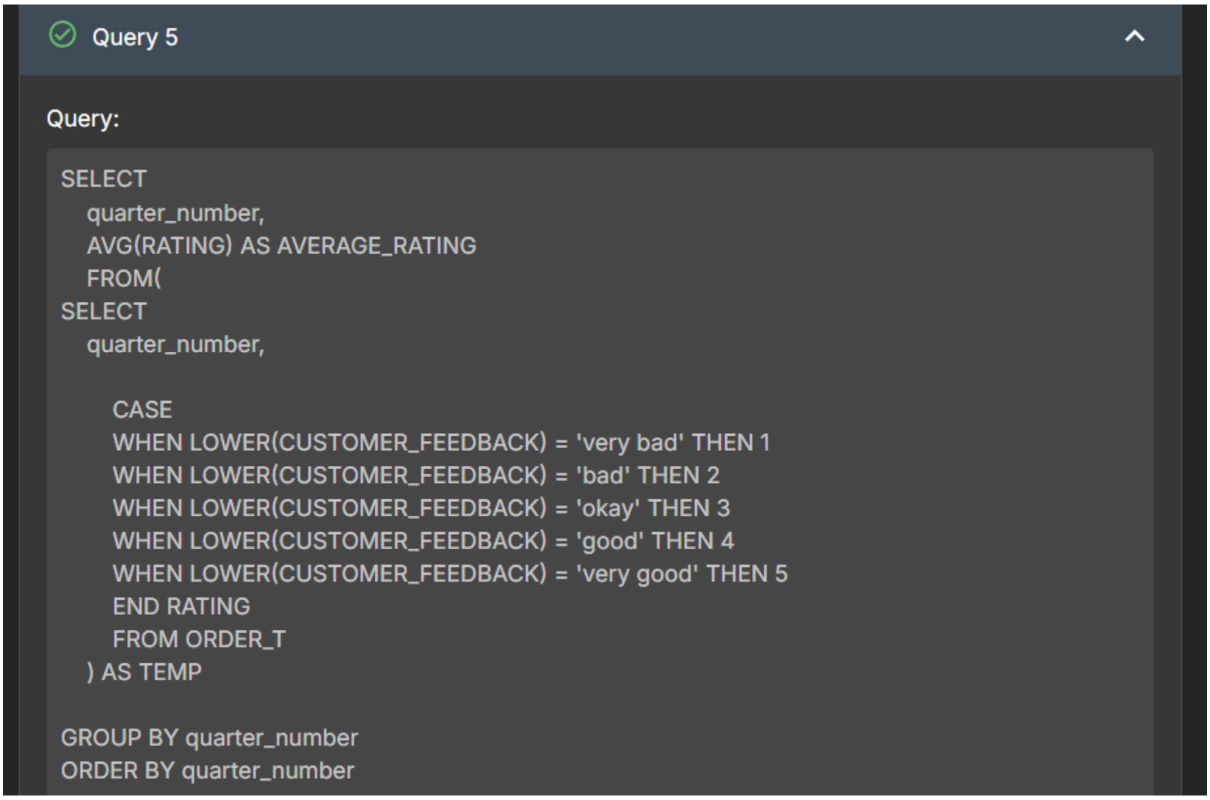
Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

SOLUTION QUERY:

```
 SELECT
    quarter_number,
    AVG(RATING) AS AVERAGE_RATING
FROM (
SELECT
    quarter_number,
     CASE
        WHEN LOWER(CUSTOMER_FEEDBACK) = 'very bad' THEN 1
        WHEN LOWER(CUSTOMER_FEEDBACK) = 'bad' THEN 2
        WHEN LOWER(CUSTOMER_FEEDBACK) = 'okay' THEN 3
        WHEN LOWER(CUSTOMER_FEEDBACK) = 'good' THEN 4
        WHEN LOWER(CUSTOMER_FEEDBACK) = 'very good' THEN 5
        END RATING
        FROM ORDER_T
    ) AS TEMP
GROUP BY quarter_number
ORDER BY quarter_number;


SELECT
          AVG(RATING) AS OVERALL_AVERAGE_RATING
FROM (
    SELECT
        CASE
```

```
    WHEN LOWER(CUSTOMER_FEEDBACK) = 'very bad' THEN 1
    WHEN LOWER(CUSTOMER_FEEDBACK) = 'bad' THEN 2
    WHEN LOWER(CUSTOMER_FEEDBACK) = 'okay' THEN 3
    WHEN LOWER(CUSTOMER_FEEDBACK) = 'good' THEN 4
    WHEN LOWER(CUSTOMER_FEEDBACK) = 'very good' THEN 5
    END RATING
    FROM ORDER_T
) AS TEMP
```



Query 5

Query:

```
SELECT
  quarter_number,
  AVG(RATING) AS AVERAGE_RATING
  FROM(
SELECT
  quarter_number,

    CASE
    WHEN LOWER(CUSTOMER_FEEDBACK) = 'very bad' THEN 1
    WHEN LOWER(CUSTOMER_FEEDBACK) = 'bad' THEN 2
    WHEN LOWER(CUSTOMER_FEEDBACK) = 'okay' THEN 3
    WHEN LOWER(CUSTOMER_FEEDBACK) = 'good' THEN 4
    WHEN LOWER(CUSTOMER_FEEDBACK) = 'very good' THEN 5
    END RATING
    FROM ORDER_T
  ) AS TEMP

GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	AVERAGE_RATING
1	3.554838709677419
2	3.354961832061069
3	2.9563318777292578
4	2.3969849246231156

Query 6

Query:

```
SELECT
 AVG(RATING) AS OVERALL_AVERAGE_RATING
FROM (
   SELECT
      CASE
      WHEN LOWER(CUSTOMER_FEEDBACK) = 'very bad' THEN 1
      WHEN LOWER(CUSTOMER_FEEDBACK) = 'bad' THEN 2
      WHEN LOWER(CUSTOMER_FEEDBACK) = 'okay' THEN 3
      WHEN LOWER(CUSTOMER_FEEDBACK) = 'good' THEN 4
      WHEN LOWER(CUSTOMER_FEEDBACK) = 'very good' THEN 5
      END RATING
   FROM ORDER_T
) AS TEMP
```

Output:

Showing 1 rows

OVERALL_AVERAGE_R...
3.135

OBSERVATION:

- Overall average rating is around 3.13
- There is continuous decline, Rating decreases steadily from Q1- Q4 indicating clear downward trends.
- Below average in later quarters Q3 and Q4 rating are neutral below 3
- Indicating negative customer experience

INSIGHT:

- Customer satisfaction is declining
- Average rating continuously decreasing across quarters indicating customer satisfaction worsening over time.
- Lowest decline in Q4 indicating sharp decline in customer experience towards the end of period
- Possible problem might be product quality, delivery/service delay or pricing dissatisfactions.
- Early warning Q1-Q2-Q3 indicating issues starts early and worsen over time.
- Overall experience is neutral to slightly negative.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

SOLUTION QUERY:

```
SELECT
        quarter_number,
        SUM (CASE WHEN LOWER(CUSTOMER_FEEDBACK) = 'very bad' THEN 1
        ELSE 0 END) *100.0 / COUNT (*) AS VERY_BAD,
        SUM (CASE WHEN LOWER(CUSTOMER_FEEDBACK) = 'bad' THEN 1
        ELSE 0 END) *100.0 / COUNT (*) AS BAD,
        SUM (CASE WHEN LOWER(CUSTOMER_FEEDBACK) = ‘okay ’THEN 1
        ELSE 0 END) *100.0 / COUNT (*) AS OKAY,
        SUM (CASE WHEN LOWER(CUSTOMER_FEEDBACK) = 'good' THEN 1
        ELSE 0 END) *100.0 / COUNT (*) AS GOOD,
        SUM (CASE WHEN LOWER(CUSTOMER_FEEDBACK) = 'very good' THEN 1
        ELSE 0 END) *100.0 / COUNT (*) AS VERY_GOOD,
FROM ORDER_T
GROUP BY quarter_number
ORDER BY quarter_number;
```

Query 7

Query:

```
SELECT
quarter_number,
   SUM(CASE WHEN LOWER(CUSTOMER_FEEDBACK) = 'very bad' THEN 1 ELSE 0 END)*100.0 /
COUNT(*) AS VERY_BAD,

   SUM(CASE WHEN LOWER(CUSTOMER_FEEDBACK) = 'bad' THEN 1 ELSE 0 END)*100.0 /
COUNT(*) AS BAD,
   SUM(CASE WHEN LOWER(CUSTOMER_FEEDBACK) = 'okay' THEN 1 ELSE 0 END)*100.0
/COUNT(*) AS OKAY,
   SUM(CASE WHEN LOWER(CUSTOMER_FEEDBACK) = 'good' THEN 1 ELSE 0 END)*
100.0/COUNT(*) AS GOOD,
   SUM(CASE WHEN LOWER(CUSTOMER_FEEDBACK) = 'very good' THEN 1 ELSE 0 END)*100.0/
COUNT(*) AS VERY_GOOD

FROM ORDER_T
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	VERY_BAD	BAD	OKAY	GOOD	VERY_GOOD
1	10.96774193548387	11.290322580645162	19.032258064516128	28.70967741935484	30
2	14.885496183206106	14.122137404580153	20.229007633587788	22.137404580152673	28.62595419847328
3	17.903930131004365	22.707423580786028	21.83406113537118	20.96069868995633	16.59388646288209
4	30.65326633165829	29.14572864321608	20.10050251256281б	10.050251256281408	10.05025125628140[illegible]

OBSERVATION

- Quarter 1: Higher % of GOOD (~28.7%) and VERY GOOD (~30%)
- Lower % of VERY BAD (~11%)
- Customer were mostly satisfied.
- Quarter 2: slightly increase in very bad (~14.85)
- Decreased in very good (~28.62)
- This indicates the early sign of declining satisfaction
- Quarter 3: very bad (~17.90) and bad (~22.70)
- Clear shift toward dissatisfaction.
- Quarter 4: VERY BAD (~30.6%) and BAD (~29.1%) peaked
- GOOD (~10%) and VERY GOOD (~10%) dropped sharply
- Majority of customers are now dissatisfied

INSIGHT:

- If the percentage of Very Bad and Bad feedback increases across quarters, it indicates that customer dissatisfaction is increasing over time. If these percentages decrease while Good and Very Good increase, then customer satisfaction is improving.

Question 6: What is the trend of the number of orders by quarter?

SOLUTION QUERY:

```
SELECT
    quarter_number,
    COUNT (order_id) AS TOTAL_ORDER


FROM ORDER_T
GROUP BY quarter_number
ORDER BY quarter_number;
```

Query 8

Query:

```
SELECT
    quarter_number,
    COUNT(order_id) AS TOTAL_ORDER

FROM ORDER_T
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	TOTAL_ORDER
1	310
2	262
3	229
4	199

OBSERVATION

- There is a consistent decline in the number of orders across each quarter.
- This shows a clear downward trend in customer orders over time.

INSIGHT:

- No quarters show recovery or growth
- There is drastic decline in orders, the biggest decline is between Q1 TO Q2 suggesting the problem may have started earlier
- Business is losing customers or orders steadily

QUESTION 7 Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

SOLUTION QUERY:

```
SELECT
quarter_number, TOTAL_REVENUE, PREV_REVENUE, ROUND ((TOTAL_REVENUE - PREV_REVENUE) * 100.0 / PREV_REVENUE, 2) AS PERCENTAGE_CHANGE
FROM (
    SELECT
    quarter_number, TOTAL_REVENUE,
    LAG(TOTAL_REVENUE) OVER (ORDER BY quarter_number)
    AS PREV_REVENUE
    FROM (
    SELECT O. quarter_number,
SUM(O.quantity * O.vehicle_price * (1 - O.discount)) AS TOTAL_REVENUE FROM ORDER_T O
    GROUP BY O.quarter_number
) AS TEMP_A
) AS TEMP_B
ORDER BY quarter_number;
```

Query:

```
SELECT
   quarter_number,
   TOTAL_REVENUE,
   PREV_REVENUE,
   ROUND((TOTAL_REVENUE - PREV_REVENUE) * 100.0 / PREV_REVENUE, 2) AS
PERCENTAGE_CHANGE
FROM (
   SELECT
      quarter_number,
      TOTAL_REVENUE,
      LAG(TOTAL_REVENUE) OVER(ORDER BY quarter_number) AS PREV_REVENUE
   FROM (
      SELECT
         O.quarter_number,
         SUM(O.quantity * O.vehicle_price * (1 - O.discount)) AS TOTAL_REVENUE
      FROM ORDER_T O
      GROUP BY O.quarter_number
   ) AS TEMP_A
) AS TEMP_B
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	TOTAL_REVENUE	PREV_REVENUE	PERCENTAGE_CHANG
1	18032549.899600018		
2	13122995.7562	18032549.899600018	-27.23
3	8882298.8449	13122995.7562	-32.32
4	8573149.280599998	8882298.8449	-3.48

OBSERVATION

- Revenue is decreasing every quarter.
- No recovery observed across the year.
- Biggest decline occurred during Quarter 3.
- Indicates a major issue during this period
- Drop reduces to - 3.48%

INSIGHT

- The company is experiencing a significant and continuous decline in net revenue, with the sharpest drop in Q3. Although the decline slows in Q4, the overall trend remains negative, indicating serious business performance issues.

Question 8: What is the trend of net revenue and orders by quarters?

Solution query:

SELECT

quarter_number, TOTAL_ORDER, NET_REVENUE,

LAG(TOTAL_ORDER) OVER (ORDER BY quarter_number) AS PREV_ORDER, LAG(NET_REVENUE) 999 1gh2112trs1 x``'a0'rOst#WVER (ORDER BY quarter_number) AS PREV_REVENUE

FROM (

SELECT quarter_number,

COUNT (order_id) AS TOTAL_ORDER,

SUM (QUANTITY * vehicle_price * (1 - discount)) AS NET_REVENUE

FROM order_t

GROUP BY quarter_number

) AS TEMP

ORDER BY quarter_number;

Query:

```
SELECT
   quarter_number,
   TOTAL_ORDER,
   NET_REVENUE,
   LAG(TOTAL_ORDER) OVER(ORDER BY quarter_number) AS PREV_ORDER,
   LAG(NET_REVENUE) OVER(ORDER BY quarter_number) AS PREV_REVENUE
FROM(
   SELECT
      quarter_number,
      COUNT(order_id) AS TOTAL_ORDER,
      SUM(QUANTITY * vehicle_price * (1 - discount)) AS NET_REVENUE
   FROM order_t
   GROUP BY quarter_number
) AS TEMP
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	TOTAL_ORDER	NET_REVENUE	PREV_ORDER	PREV_REVEN
1	310	18032549.899600018		
2	262	13122995.7562	310	18032549.89
3	229	8882298.8449	262	13122995.75
4	199	8573149.280599998	229	8882298.84

OBSERVATION:

- Both Revenue and Orders are Declining
- Orders drop from 310 to 199
- Revenue drops from 18M → 8.5M
- Indicates a consistent downward business trend
- Strong Correlation Between Orders & Revenue
- As the number of order decreases revenue also decreases proportionally
- Less order trends to less revenue
- Q3 and Q4 shows the biggest drop in both the metrics
- Suggests a major operational or customer-related issue during this period
- Possible early signs of stabilization, but still negative

INSIGHT:

- Customer dissatisfaction increases orders and revenue are in downwards
- Poor customer experience → Fewer orders → Declining revenue
- All are interconnected to each other
- The company is experiencing a consistent decline in both orders and net revenue across all quarters, indicating worsening business performance

QUESTION 9: What is the average discount offered for different types of credit cards?

SOLUTION QUERY:

SELECT C. credit_card_type,

AVG (O. discount) AS AVG_DISCOUNT

FROM customer_t C

JOIN ORDER_T O ON O.customer_id = C.customer_id

GROUP BY C.credit_card_type

ORDER BY C. credit_card_type;

```
SELECT C.credit_card_type,
   AVG(O.discount) AS AVG_DISCOUNT

FROM customer_t C
JOIN ORDER_T O ON O.customer_id = C.customer_id
GROUP BY  C.credit_card_type
ORDER BY C.credit_card_type
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	AVG_DISCOUNT
americanexpress	0.616326530612245
bankcard	0.6095454545454548
china-unionpay	0.6221739130434784
diners-club-carte-blan...	0.6144897959183674
diners-club-enroute	0.5997916666666666
diners-club-international	0.584
diners-club-us-ca	0.6146153846153846
instapayment	0.620625
jcb	0.6073820754716984
laser	0.643846153846154

OBSERVATION:

- Laser card (~64.38%) offers the highest average discount.
- Diners Club International (~58.40%) has the lowest discount.
- Most cards fall in a narrow range of 60%–62%
- Indicates consistent discount strategy across payment methods

INSIGHT:

- The company is offering almost uniform discounts across credit card types, with minor variations.
- The average discount is fairly consistent across credit card types (around 60–62%), with Laser offering the highest discount and Diners Club International the lowest. This suggests a balanced discount strategy with slight promotional differences across payment providers.

QUESTION 10: What is the average time taken to ship the placed orders for each quarter?

SOLUTION QUERY:

SELECT

quarter_number,

AVG (julianday(ship_date) - julianday(order_date)) AS AVG_SHIPPING_TIME

FROM ORDER_T

GROUP BY quarter_number

ORDER BY quarter_number;

```
SELECT
  quarter_number,
  julianday(ship_date) - julianday(order_date) AS AVG_SHIPPING_TIME

FROM ORDER_T
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	AVG_SHIPPING_TIME
1	65
2	123
3	46
4	35

OBSERVATION:

- Consistent Increase in Shipping Time
- Shipping time increases every quarter:
- Significant Delay in Later Quarters

- Shipping time more than triples from Q1 (~57) to Q4 (~174)
- Sharpest Increase After Q2
- Major jump observed in Q3 and Q4

INSIGHT:

- The average shipping time is steadily increasing across quarters, indicating worsening operational efficiency. This delay is likely contributing to declining customer satisfaction, fewer orders, and reduced revenue.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
48610993.78	1000	994	3.14
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
8573149.26	199	97.96	44.1

INSIGHT

- Customer Satisfaction is Declining bad and very bad is increasing over quarter.
- Positive feedbacks dropped to ~20% in Q4.
- Customers are increasingly unhappy with the service
- Orders and Revenue are Falling Together. Orders: 310 → 199, Revenue: 18M → 8.5M
- Shipping Time is the Major Issue, Shipping time increased drastically: 57 → 174 days
- This is likely the root cause of dissatisfaction
- Critical Period: Q2–Q3 Sharp drops in revenue and satisfaction start here

BUSINESS RECOMMENDATION

- Improve Logistics & Delivery Time (Top Priority)
- Optimize supply chain and warehouse operations
- Tie up parters for faster delivery
- Implement real-time order tracking
- Reduce shipping time from 174 → <70 days
- Provide proactive updates on delays
- Improve customer support response time
- This helps rebuild trust and reduce negative feedback
- Maintain optimal stock levels
- Avoid delays due to stockouts
- we need to use forecasting model to predict demands
- we can implement smart discount strategy already higher discount of around 60% are already given its better we can focus on service improvement.

- Offer targeted promotions to dissatisfied customers
- Discounts alone won’t fix the problem
- Main Root Cause Analysis (Q2–Q3 Drop)
- So, we need to investigate supplier issue, delivery partner performances.
- Collect feedback regularly this will Helps recover lost customers The business decline is primarily driven by increasing shipping delays, which led to customer dissatisfaction, reduced orders, and falling revenue.